IP Monitor & Blocker – Documentation

Purpose

This script is designed to **monitor a local network (e.g., 192.168.1.x)**, automatically block unauthorized IPs using Windows Firewall, and send alerts via Telegram or PLC.
It supports **manual blocking** (which cannot be auto-unblocked) and **automatic blocking** based on configurable IP ranges.

Main Features

- **Automatic IP Blocking:**
  Blocks IPs outside the allowed range or not in the always-allowed list.
- **Manual IP Blocking:**
  Allows the user to manually block any IP, even if it's in the allowed range.
- **Auto-Unblock:**
  Automatically unblocks IPs that become allowed (unless manually blocked).
- **Network Scanning:**
  Uses ARP and ICMP to detect active devices.
- **Packet Monitoring:**
  Monitors live packets to catch new devices instantly.
- **Alerts:**
  Sends notifications via Telegram and/or writes to a PLC.
- **Persistent Block Lists:**
  Tracks blocked IPs in blocked_ips.json and manual blocks in manual_blocked_ips.json.
- **Configurable:**
  Reads allowed IP range, prefix, and always-allowed IPs from ip_monitor_config.json.

How It Works

1. Configuration

- **Config File:**
  ip_monitor_config.json
  Contains:
  - ALLOWED_PREFIX (e.g., "192.168.1.")
  - ALLOWED_START (e.g., 1)
  - ALLOWED_END (e.g., 245)
  - ALWAYS_ALLOWED_IPS (list of IPs always allowed)
  - ALERT_MODE ("Online" for Telegram, "Offline" for PLC)

2. Block Lists

- **blocked_ips.json:**
  All currently blocked IPs (auto and manual).
- **manual_blocked_ips.json:**
  IPs manually blocked by the user (never auto-unblocked).

3. Network Scanning

- **ARP Scan:**
  Finds active devices in the subnet.
- **ICMP Scan:**
  Pings all possible IPs in the subnet to find devices not responding to ARP.

4. Blocking Logic

- **Auto-block:**
  If an IP is detected and is not allowed (not in range, not always allowed), it is blocked and added to blocked_ips.json.
- **Manual block:**
  If the user blocks an IP manually (via GUI/app), it is added to both blocked_ips.json and manual_blocked_ips.json.

5. Unblocking Logic

- **Auto-unblock:**
  If an IP is blocked but becomes allowed (due to config change), it is auto-unblocked **unless it is in manual_blocked_ips.json**.
- **Manual unblock:**
  Only possible via the GUI/app for IPs in manual_blocked_ips.json.

6. Packet Monitoring

- **Live packet sniffing:**
  Monitors all IP and ARP packets.
  - If a new device is detected, applies block logic.
  - If a blocked device becomes allowed, applies unblock logic (skips manual blocks).

7. Alerts

- **Telegram:**
  Sends messages for block/unblock events.
- **PLC:**
  Writes block events to a PLC if in "Offline" mode.

Key Functions

- **get_live_config():**
  Loads and caches the config file.
- **load_blocked_ips()/save_blocked_ips():**
  Loads/saves the set of blocked IPs.
- **load_manual_blocked_ips()/save_manual_blocked_ips():**
  Loads/saves the set of manually blocked IPs.
- **block_ip(ip):**
  Adds Windows Firewall rules to block the IP.

- **unblock_ip(ip):**
  Removes Windows Firewall rules for the IP.
- **packet_callback(packet):**
  Handles live packets, applies block/unblock logic.
- **active_arp_icmp_scan_and_block():**
  Scans the network and applies block logic.
- **unblock_ips_now_allowed():**
  Unblocks IPs that are now allowed (skips manual blocks).
- **periodic_scan():**
  Runs ARP/ICMP scan and auto-unblock every 3 seconds.

How to Use

1. **Configure Allowed Range:**
   Edit `ip_monitor_config.json` to set allowed prefix, start, end, and always-allowed IPs.
2. **Run the Script:**
   Start Sms_Alert.py as administrator.
3. **Manual Block/Unblock:**
   Use the GUI/app to manually block or unblock IPs.
4. **Monitor:**
   The script will automatically block unauthorized IPs and send alerts.

Important Notes

- **Manual blocks are persistent:**
  IPs in `manual_blocked_ips.json` will NOT be auto-unblocked, even if they become allowed by config.
- **Auto blocks are dynamic:**
  IPs blocked automatically will be unblocked if they become allowed.
- **File paths:**
  All JSON files must be in the same directory as the script/app.
- **Run as administrator:**
  Required for firewall rule changes.

Typical Workflow

1. **Start with a range (e.g., 1–200):**
   Only IPs in `192.168.1.1` to `192.168.1.200` are allowed.
2. **Expand range (e.g., 1–255):**
   Previously blocked in-range IPs are auto-unblocked (unless manually blocked).
3. **Manual block:**
   Block any IP manually; it stays blocked until manually unblocked.
4. **Change config:**
   Script/app respects new config and updates blocks/unblocks accordingly.

Troubleshooting

- **Manual block is auto-unblocked:**
  Check that both app and script skip manual blocks in their auto-unblock logic and use the same manual_blocked_ips.json file.
- **Blocked IPs not updating:**
  Ensure you run as administrator and all JSON files are in the correct directory.